

Article

Impact of IoT (BLE) and LTE on Fleet Efficiency: An Empirical Case Study of a Bulgarian Logistics Company

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Abstract

Background: Despite the widespread adoption of IoT and LTE technologies in logistics, empirical research quantifying the economic impact of preventing operational errors remains scarce. **Methods:** This paper presents a case study of a Bulgarian transport company with over 800 tractors and 650 trailers, awarded "Transport Company of the Year 2025" at the Plovdiv Business Awards. An architecture combining BLE sensors, LTE trackers, and two-way integration between the TMS and the Kitin GPS platform was deployed. Using a mixed-methods approach (quantitative KPIs and qualitative process analysis), we measured the effect on annual excess mileage, trailer swap registration time, and error frequency. **Results:** The results show a 90% reduction in excess mileage (from 10,000 km to 1000 km), a 99.9% faster swap registration time, and a 90% reduction in errors. Annual savings amount to €19,750, with an investment payback period of 13–14 months (334% ROI over 5 years). **Conclusions:** The scientific contribution is the empirical demonstration that error prevention (correct trailer attachment) dominates asset tracking as a source of value. Theoretical and practical recommendations for fleet managers are presented.

Keywords: fleet efficiency; cost reduction; KPI; TMS; GPS; Kitin; BLE; LTE; error prevention; ROI; empirical research

Academic Editors: Artur Kierzkowski, Tomasz Nowakowski, Agnieszka A. Tubis, Franciszek Restel, Tomasz Kisiel, Anna Jodejko-Pietruczuk, Mateusz Zając, Viktoria Ivannikova, Michał Stosiak and Andrija Vidović

Received: 3 July 2026

Revised: 2 September 2026

Accepted: 3 September 2026

Published: 9 September 2026

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1. Introduction

The logistics sector faces increasing pressure from multiple directions. On the one hand, fuel prices remain high (the weighted average price of diesel in the European Union in August 2025 was approximately €1.54 per liter [1]), making every kilometer saved critical for profitability. On the other hand, profit margins in transport are often below 3%, and customers demand ever-higher accuracy and real-time visibility over their shipments. In this context, every delay, every excess kilometer, and every error in fleet management directly impacts the company's financial performance.

Modern fleet management increasingly relies on digital solutions and telematics. The global Transportation Management System (TMS) market was valued at approximately \$15 billion in 2025 and is expected to reach \$40.3 billion by 2035, with a compound annual growth rate of 10.6% [2]. Other sources cite even higher values: \$18.7 billion in 2025 with a forecast of \$44.8 billion by 2034. This growth is driven primarily by the need for planning, execution, route optimization, freight management, and real-time analytics. However,

despite significant software investments, one of the greatest challenges remains physical asset tracking, particularly trailers, which often remain “blind” to the system when disconnected from tractors.

Traditional trailer tracking solutions rely on two main approaches. The first is manual entry of identification numbers by dispatchers into the TMS, a method prone to errors, delays, and incomplete data. The second is installing active GPS trackers on every trailer [3]. The cost of a basic GPS tracking system ranges from \$100 to \$300 per device, but real costs are significantly higher, as each device requires a monthly SIM card fee, regular maintenance, and battery replacement. Furthermore, pairing a GPS tracker with a specific tractor and the TMS is often a cumbersome and expensive process, requiring specialized software and integration time. For a large fleet of over 800 tractors and 650 trailers, these costs and operational difficulties become prohibitive.

An alternative approach gaining traction is the use of BLE (Bluetooth Low Energy) beacons [4] or Queclink WID330 [5]. These are small, low-cost devices with a battery life of 2 to 3 years, requiring no SIM cards or monthly fees. In combination with LTE/5G trackers in tractors, BLE beacons enable automatic detection of an attached trailer, data transmission to a GPS platform, and two-way communication with the TMS. This approach eliminates the need for expensive GPS devices on every trailer, simplifies pairing, and drastically reduces operational costs. However, systematic empirical studies quantifying the economic impact of such an architecture in a large-scale fleet are still lacking.

This study aims to fill this gap by presenting a real-world case study of a Bulgarian logistics company with over 800 tractors and 650 trailers that deployed a BLE/LTE architecture integrated with a TMS. This study addresses the following research questions:

- RQ1: To what extent does the integration of BLE sensors, LTE trackers, and a two-way TMS–GPS platform affect operational KPIs (excess mileage, swap registration time, attachment errors)?
- RQ2: What is the economic return (ROI) and payback period of such architecture for a large-scale fleet (>800 tractors)?
- RQ3: Based on the case study evidence, what is the relative value of error prevention compared to asset tracking, and what are the practical implications for fleet managers? We propose that error prevention generates higher operational and financial returns than asset tracking, and we support this proposition through a comparative cost–benefit analysis and the quantified outcomes of the implemented BLE/LTE architecture.

This paper is structured as follows. Following this introduction, Section 2 presents a literature review and theoretical framework linking this study to quality cost theory, lean logistics, and proactive management. Section 3 describes the methodology, including the case study design, data collection, and the deployed solution architecture. Section 4 presents the results: operational and financial KPIs before and after implementation. Section 5 discusses the results, compares them with other studies, and outlines theoretical and practical implications. Section 6 presents the research contributions. Section 7 outlines limitations and directions for future research. Section 8 concludes with a summary of the main findings and recommendations for fleet managers.

2. Literature Review and Theoretical Framework

Research on IoT in logistics can be grouped into three main directions. Recent academic research has extensively explored the application of IoT technologies in logistics, with studies examining the transition from real-time visibility to operational benefits, the development of structured decision-making models for digital technology

implementation, and the integration of IoT telemetry with cloud analytics for fleet automation. Furthermore, BLE-based real-time location systems have been investigated for internal logistics operations, and multi-radio devices have been developed for container monitoring and tracking. The use of IoT-enabled BLE systems for fleet tracking has also been proposed as an alternative to traditional GPS-based approaches. Big Data Analytics and IoT in logistics have demonstrated significant improvements in driver safety, cost reduction, and environmental performance, while data-driven approaches using multi-objective optimization have been applied to identify critical parameters for fleet performance monitoring. First, there has been work focusing on TMS optimization and digital infrastructure [6,7]. Second, there are studies on sensor networks and real-time data processing [8,9]. Third, there have been applications of 5G/LTE technologies in transport corridors [10,11]. Common to all is the focus on visibility: where an asset is located. Only recently has attention turned to error prevention [12,13] but without quantifiable economic effects. Academic research in leading logistics journals has examined the broader implications of IoT adoption in fleet management. Explored the transition from real-time visibility to operational benefits, highlighting the tensions that emerge on this path. Developed a structured decision-making model for implementing digital technologies in the logistics sector, while case studies on Big Data Analytics and IoT in logistics have demonstrated significant improvements in driver safety, cost reduction, and environmental performance. Data-driven approaches to fleet management have also been advanced, with studies employing multi-objective optimization methods to identify critical parameters for fleet performance monitoring. To structure our analysis, we apply three established theoretical concepts. First, the theory of quality costs [14,15] distinguishes prevention costs (preventing errors) from failure costs (correcting errors after they occur). In a logistics context, this means that investments in systems to prevent incorrect trailer attachments should yield higher returns than post-factum tracking systems. Our study provides empirical support for this proposition. Second, lean logistics and the zero-defect mindset [16,17] emphasize waste elimination. In our case, a 90% reduction in wrong attachments eliminates a significant source of waste: excess mileage, overtime, and penalties. This demonstrates that IoT-based prevention is a powerful tool for achieving lean goals in fleet operations. Third, the supply chain management literature distinguishes proactive (prevention) from reactive (correction) approaches [18,19]. Our study shows that, in the context of fleet management, the proactive BLE/LTE approach yields significantly higher economic benefits than reactive tracking systems. Our theoretical framework distinguishes two types of value: (a) recovery value—locating lost assets, reducing theft; (b) prevention value—avoiding errors before they occur. We hypothesize that for fleets with high trailer swap frequency (≥ 100 per week), prevention dominates.

3. Methodology

A single-case study design [20] with embedded quantitative and qualitative analysis was used. The study stages included the following:

- **Sample and context:** A Bulgarian logistics company with over 800 tractors, 650 trailers, and an average of 100 weekly trailer swaps. In 2025, the company received the “Transport Company of the Year” award at the Plovdiv Business Awards, with the jury citing “impressive results from digital transformation and investments in innovative fleet management systems.” The study period covered 12 months before and 12 months after implementation.
- **Data collection:** Data were collected from two main sources: (1) the TMS—providing data on planned vs. actual routes, trailer swap registration time, and the number of wrong attachment incidents; (2) the Kitin GPS platform—providing real-time

location data for tractors and trailers, as well as timestamps for each attachment/detachment. Data were collected automatically, ensuring objectivity.

- **Intervention:** The deployed architecture included: BLE beacons (e.g., Teltonika EYE BTSID1 [4]; LTE trackers (Ruptela HCV5) in tractors [21]; two-way integration between the Kitin GPS platform and the TMS. The cycle from attachment to route visualization takes less than 2 s.
- **Data analysis:** Descriptive statistics (means, percentages, standard deviations) were used to compare KPIs between the baseline and evaluation periods. Both absolute change (in kilometers, seconds, number of incidents) and relative change (in percentages) were calculated. Financial analysis included net present savings and payback period calculations based on annual savings versus one-time investment. ROI was calculated as $(\text{net savings over 5 years} \div \text{investment}) \times 100\%$.

Before implementation, the company relied on manual entry of the composition (tractor + trailer) by the dispatcher managing the fleet of tractors and drivers. This process took between 5 and 10 min per swap and was prone to errors, such as incorrect trailer number entry, delayed system updates, or mismatches between planned and actual attachments. The dispatcher had to manually verify available trailers, communicate with drivers by phone, and update the TMS, leading to time waste and potential errors. A summary of the company's key operational parameters is provided in Table 1.

Table 1. Company profile.

Parameter	Data
Tractors	>800
Trailers	~650
Weekly trailer swaps	~100
Routes	Bulgaria + Western Europe
Previous network	2G (outdated, slow)
Award	“Transport Company of the Year 2025” (Plovdiv Business Awards)—the jury cited digital transformation and innovation as key factors

The Kitin GPS platform provides several key features that support the integration, as shown in Table 2.

Table 2. Key features of the Kitin platform.

Feature	Business Benefit
Two-way API	TMS receives accurate data and returns a route
Scalable BLE processing	Works with 650+ trailers
LTE readiness	Latency < 100 ms
User-friendly interface	Zero driver training required

3.1. Data Collection and Processing

To ensure reproducibility, we provide detailed information on the data collection and processing procedures.

Study period: The baseline period (pre-implementation) covers 1 January 2024 to 31 December 2024. The intervention (system implementation) took place on 1 January 2025. The evaluation period (post-implementation) covers 1 January 2025 to 31 December 2025. This 12-month pre/post design allows for seasonal variation control and provides a sufficient sample size for statistical analysis.

Fleet growth and operational volume: During the study period, the company experienced significant fleet growth. At the start of the baseline period (January 2024), the fleet consisted of 700 tractors and 600 trailers. By the end of the evaluation period (December 2025), the fleet had expanded to 800 tractors and 650 trailers, reflecting the company's growth and market expansion. This growth resulted in a 15% increase in weekly trailer swaps, from approximately 100 per week during the baseline period to approximately 115 per week during the evaluation period. At the time of writing (2026), the fleet continues to grow and now exceeds 800 tractors and 650 trailers, confirming the scalability of the deployed solution.

Sample size and unit of analysis: During the baseline period, a total of 5240 trailer swaps were recorded (mean = 100.8 per week, SD = 8.2). During the evaluation period, a total of 5980 trailer swaps were recorded (mean = 115.0 per week, SD = 9.1). The 15% increase in swap frequency reflects the fleet growth during the study period. The unit of analysis is each individual trailer swap event. A swap event is defined as the physical detachment of a trailer from one tractor and its subsequent attachment to another tractor.

Incorrect attachment incidents: These are defined as events where the recorded trailer ID did not match the trailer physically attached to the tractor. During the baseline period (manual dispatcher entry), we identified 142 such incidents (mean = 2.73 per week, SD = 1.1). During the evaluation period (automated BLE/LTE detection), we identified 11 such incidents (mean = 0.21 per week, SD = 0.4). The 11 incidents recorded during the evaluation period were not attributable to dispatcher errors, as dispatcher entry was fully automated. Instead, these incidents resulted from drivers not using the system and from delayed notification via alternative communication channels regarding correct delivery addresses. All incidents were verified by cross-checking dispatcher records with driver confirmation and GPS tracking data.

Data sources: All data were extracted from two primary sources: (1) the TMS system logs, which record dispatcher entries, route assignments, and timestamps; and (2) the Kitin GPS platform, which records BLE detection events, GPS coordinates, and data transmission timestamps. Both systems operate independently, allowing cross-validation of key events.

Data cleaning and missing data treatment: Data were extracted through automated API queries, with no manual data entry during the extraction process. We identified no missing data in the TMS logs, as the system automatically logs every dispatcher action. In the GPS platform, we identified 0.3% missing BLE detection events (approximately 15 events per month), primarily due to temporary BLE signal loss in areas with high metallic interference. These missing events were excluded from the analysis, as they represented less than 1% of the total dataset and were randomly distributed, posing no systematic bias. No imputation methods were applied.

System integration and testing: The BLE sensor installation, system integration, and testing were conducted over a three-month period (October–December 2024) in a dedicated test environment. This testing phase did not affect the baseline or evaluation periods. The integration involved hardware installation on all trailers, configuration of the Ruptela HCV5 trackers for BLE detection, and development of the two-way API integration between the Kitin GPS platform and the TMS. Staff training was not required, as the automated process eliminated manual data entry.

KPI calculation formulas: Each KPI was calculated using the following formulas:

Annual excess mileage (km): Sum of (planned route distance–actual route distance) for all completed trips where actual distance exceeded planned distance by more than 5%. Only trips with a positive deviation greater than 5% were included, to exclude minor GPS-based fluctuations. Data were extracted from the TMS route log.

Trailer swap registration time (seconds): The time elapsed between the physical attachment of the trailer (as recorded by the Ruptela HCV5 tracker's BLE detection timestamp) and the confirmation of the swap in the TMS (as recorded by the dispatcher's entry or, in the post-implementation period, the automatic API update timestamp).

Wrong attachment incidents (per week): The number of dispatcher entries (baseline) or automatic detections (post-implementation) where the recorded trailer ID did not match the trailer physically attached to the tractor. Physical verification was conducted by cross-referencing dispatcher records with GPS tracking data and driver confirmation.

Driver distraction score: Measured as the average number of stops per shift where the driver exited the cab for ID verification purposes (baseline) vs. zero (post-implementation, as no manual verification is required). Data were collected from driver self-reporting logs (baseline) and confirmed through GPS stop duration analysis.

Address accuracy (%): The percentage of trips where the destination address entered in the TMS matched the address confirmed by the driver upon arrival. In the baseline period, address accuracy was based on dispatcher entry; in the post-implementation period, it was based on TMS-generated addresses (no manual entry).

3.2. Statistical Analysis

To quantify the statistical significance of the observed improvements, we performed comparative statistical tests between the baseline and evaluation periods. For each KPI, we calculated the mean, standard deviation (SD), and 95% confidence interval (CI). The normality of the data was assessed using the Shapiro–Wilk test. As the data were not normally distributed, we applied the non-parametric Mann–Whitney U test (also known as the Wilcoxon rank-sum test) to compare the baseline and evaluation periods. This test is appropriate for comparing two independent samples when the data are not normally distributed. Statistical significance was set at $p < 0.05$. All statistical analyses were performed using Python (SciPy package, version 1.12.0).

4. Results

The results are structured into two main groups: operational KPIs (directly related to fleet efficiency) and financial KPIs (economic benefits of the deployed architecture). The data cover a period of 12 months before and 12 months after the BLE/LTE system deployment. All values are based on automatically collected data from the TMS and the Kitin GPS platform. The operational KPIs before and after implementation are presented in Table 3.

Table 3. Operational KPIs before and after implementation.

KPI	Before (Manual)	After (BLE + LTE + TMS)	Change
Annual excess mileage	10,000 km	~1000 km	−90%
Trailer swap registration time	5–10 min	<1 s	−99.9%
Wrong attachments (weekly)	2–3	~0.2	−90%
Driver distraction	High	Eliminated	−100%
Address accuracy	85%	99%	+14%

Source: Internal TMS and Kitin GPS platform data, period 2024–2025.

The financial impact is summarized in Table 4, which shows the cost savings across different categories.

Table 4. Financial KPIs before and after implementation (EUR).

KPI	Before	After	Saved
Fuel	7500	750	6750
Driver overtime	~5000	~500	4500
Dispatcher labor	~8000	~500	7500
Customer penalties	~3000	~200	2800
Total	~23,500	~1950	~21,550

Source: Internal financial reports and TMS data, period 2024–2025. The total savings including avoided customer penalties amount to €21,550. The conservative estimate of €19,750 (excluding penalties) is used for ROI and payback calculations.

The ROI analysis presented in this study considers both hardware investment and full implementation costs[22]. While the primary hardware investment consists of BLE beacons, additional costs include software integration, installation, and maintenance.

The installation of BLE beacons is minimal due to the wireless nature of the technology. The beacons are lightweight, waterproof (IP67/IP68), and dustproof, allowing flexible mounting on any trailer surface without requiring specialized tools. They can be attached using double-sided adhesive tape or cable ties, eliminating the need for drilling, wiring, or structural modifications. This significantly reduces installation time and labor costs compared to wired GPS tracking systems.

Maintenance is equally straightforward. Battery replacement requires only a Phillips screwdriver to open the protective cover, with replacement batteries costing between €1 and €2 each. Given the 2–3-year battery life, the annual maintenance cost per beacon is less than €1, making it negligible compared to active GPS trackers that require SIM cards and monthly fees.

Regarding software integration, the company invested a one-time cost of €1000 to optimize the integration between the TMS and the Kitin GPS platform. No additional hardware or middleware was required, as the existing API infrastructure was used. Staff training was not necessary, as the automated BLE detection eliminates manual data entry, meaning dispatchers and drivers continue using the same TMS interface without additional learning effort.

To provide a complete overview of the investment, Table 5 presents a breakdown of all costs, distinguishing between hardware investment and full implementation costs.

Table 5. Full investment breakdown (hardware vs. implementation cost).

Cost Category	Item	Cost (€)	Notes
Hardware Investment	BLE beacons (650 × €35)	22,750	One-time purchase; 2–3-year battery life
Hardware Investment	Replacement batteries (650 × €2 every 2 years)	650	Annualized cost over 2 years
Implementation Cost	Software integration (TMS ↔ Kitin API)	1000	One-time cost
Implementation Cost	Installation (self-installation by staff)	0	No external labor; adhesive tape/cable ties
Implementation Cost	Staff training	0	No training required; automated process
Implementation Cost	Maintenance (battery replacement)	325	Annual cost (€0.50 per beacon per year)
Total Hardware Investment		22,750	BLE beacons only
Total Implementation Cost		1325	Software + maintenance (first year)
Total One-Time Investment		22,750	Hardware (excluding consumables)
Total Full Implementation Cost (first year)		24,075	Hardware + integration + maintenance

The annual operational savings of €19,750 (presented in Table 6) are recovered in approximately 14 months, resulting in the ROI shown in Table 7.

Table 6. Annual savings.

Category	Savings (€)
Fuel	6750
Driver overtime	5000
Dispatcher labor	8000
Total	19,750

Table 7. Return on investment.

Indicator	Value
Investment	€22,750
Annual savings	€19,750
Payback period	~13–14 months
Net savings over 5 years	€76,000
5-year ROI	334%

5. Discussion

The results confirm the three research questions and provide empirical evidence for the advantages of the BLE/LTE architecture over traditional trailer tracking solutions.

5.1. Comparison with Traditional GPS Solutions

The traditional approach to trailer tracking relies on installing active GPS trackers on every trailer. While this provides independent tracking, it is associated with significant costs and operational difficulties. A 2025 market study shows that monthly fees for GPS tracking of a single vehicle range from \$10 to \$45 [23], with basic plans starting at around \$16–\$20 per month per trailer. For a fleet of 650 trailers, this means that monthly costs are between \$6500 and \$29,250, or between \$78,000 and \$351,000 annually, which are sums that are prohibitive for many companies.

Furthermore, the initial hardware investment ranges from \$20 to \$300 per device, with more complex fleet systems costing over £500. For 650 trailers, this means a one-time cost between \$13,000 and \$195,000 just for purchasing the devices, excluding installation, maintenance, and battery replacement costs.

BLE beacons (as used in this study) offer a substantial alternative. At approximately €35 per device and no monthly fees, the total investment for 650 trailers is €22,750, an amount that is recovered in approximately 14 months from operational savings alone (as shown in Table 7). This makes the BLE solution between 3 and 15 times cheaper in the long term compared to active GPS trackers.

The statistical significance of the improvements was assessed using the Mann–Whitney U test. The detailed results, including means, standard deviations, confidence intervals, and p-values, are shown in Table 8.

Table 8. Statistical comparison of key operational KPIs before and after implementation.

KPI	Period	Mean	SD	95% CI	p-Value	Significance
Excess mileage (km/year)	Baseline	10,000	850	(9150–10,850)	<0.001	***
Excess mileage (km/year)	Evaluation	1000	120	(880–1120)	<0.001	***
Swap registration time (seconds)	Baseline	450	120	(330–570)	<0.001	***

Swap registration time (seconds)	Evaluation	0.8	0.2	(0.6–1.0)	<0.001	***
Wrong attachments (per week)	Baseline	2.73	1.10	(1.63–3.83)	<0.001	***
Wrong attachments (per week)	Evaluation	0.21	0.40	(−0.19–0.61)	<0.001	***
Driver distraction (stops per shift)	Baseline	2.50	0.80	(1.70–3.30)	<0.001	***
Driver distraction (stops per shift)	Evaluation	0.00	0.00	(0.00–0.00)	<0.001	***
Address accuracy (%)	Baseline	85.0	8.5	(76.5–93.5)	<0.001	***
Address accuracy (%)	Evaluation	99.0	1.5	(97.5–100.5)	<0.001	***

Note: *** $p < 0.001$.

5.2. Pairing and Integration Challenges.

One of the most significant operational issues with traditional GPS systems is pairing a tracker with a specific tractor and the TMS. In most cases, this requires manual intervention: the dispatcher must enter the trailer number into the system, link it to the specific tractor, and confirm the composition. This process is prone to errors, delays, and mismatches between planned and actual attachments.

In the present case, manual entry of the composition took between 5 and 10 min per swap. With 100 weekly trailer swaps, this meant between 500 and 1000 min per week (or 8–16 h) of administrative work. After the BLE/LTE architecture deployment, this process was reduced to under 1 s—automatic trailer detection with no human intervention. This not only saves dispatcher time but also eliminates errors caused by manual entry.

Integrating GPS trackers with the TMS is also challenging. It often requires API or EDI connections that demand specialized knowledge and setup time. In our case, the Kitin GPS platform already had a two-way API, making integration seamless and cost-free. This highlights the importance of choosing a platform that supports open standards and easy integration with existing TMSs. A summary of how decision-making changed across different operational situations is provided in Table 9.

Table 9. Comparison of decision making before and after implementation.

Situation	Before (Manual/Batch)	After (Real-Time)	Effect
Wrong trailer attached	Discovered after 20–30 km	Discovered before departure	Zero excess km
Customer changes address	Driver stops, calls dispatcher	TMS generates new route in <2 s	Flexibility, satisfaction
Cross-border operations	Roaming, delays	Seamless LTE operation	Continuous visibility

5.3. Comparison with Other Studies

Our results are consistent with earlier studies showing that IoT-based solutions can yield significant logistics savings. Westernacher [24] (2025) documented a 98% KPI match after SAP TMS implementation, while Samsara [25] (2025) reported a 16% annual cost reduction. Our results (90% excess mileage reduction and 99.9% faster swap registration) are at the high end of this range, which is attributed to the focus on error prevention rather than just route optimization.

Reducing reaction time from minutes to seconds leads to an exponential decrease in subsequent error correction costs. Our results (99.9% faster swap registration) confirm this relationship and demonstrate that two-way communication between the TMS and GPS

platform is a key enabler of these outcomes. Danach et al. [9] present a model for adaptive real-time decision making based on IoT. Our study complements this model by showing how real-time (sub-second latency) enables not only correction but also prevention of errors before they occur.

Comparative Cost–Benefit Analysis

To address the question of whether error prevention dominates asset tracking as a source of value, we developed a comparative cost–benefit model based on the following assumptions:

- Asset recovery value (GPS tracking): Based on industry data, the average annual value of recovering lost or misplaced trailers using active GPS tracking is estimated at approximately €5000–€10,000 per 100 trailers [26,27]. For a fleet of 650 trailers, this translates to an estimated recovery value of €32,500–€65,000 annually, assuming 10–20 lost trailer incidents per year.
- Error prevention value (BLE/LTE): Based on the quantified results of this study, the annual savings from preventing wrong trailer attachments, excess mileage, dispatcher labor, and driver overtime amount to €19,750 (Table 6). This figure represents the direct financial benefit of error prevention.
- Cost comparison: The total annual cost of operating a BLE-based system (hardware amortization + maintenance) is approximately €1325, compared to an estimated annual cost of €78,000–€351,000 for active GPS tracking (monthly fees only, excluding hardware and maintenance) [23].

5.4. Future Potential of the BLE Ecosystem

Although this study focuses on trailer tracking and error prevention, BLE technology offers significantly greater potential for fleet optimization. The same BLE beacons used for trailer identification can be extended with additional sensors without changing the core architecture. This includes temperature sensors for cold chain monitoring, tire pressure and temperature sensors (TPMSs) which can reduce fuel consumption by up to 3–5% [28] and improve safety [29] and prevent roadside assistance and accidents, and sensors for humidity, door opening, and shock/vibration.

5.5. Theoretical and Practical Implications

Theoretical implications: Our study enriches quality cost theory [14,15] in the logistics context. While traditional applications of this theory focus on manufacturing processes, we demonstrate that prevention costs (investments in BLE/LTE architecture) are not only lower than failure costs (excess mileage, penalties, lost time) but also yield measurable and rapid returns. This confirms the hypothesis that error prevention dominates asset tracking as a source of value in logistics.

Practical implications: For fleet managers, our study provides clear guidance:

1. BLE beacons are more cost-effective than active GPS trackers for fleets with over 200 trailers, due to the absence of monthly fees and lower hardware costs.
2. Two-way integration between the TMS and GPS platform is critical: one-way data flow (GPS → TMS) provides visibility but does not prevent routing errors.
3. A pilot deployment with 50–100 trailers is recommended, followed by scaling, to validate results in the specific context of the company.

The managerial recommendations presented in this study are based on the specific case of a large Bulgarian logistics company with over 800 tractors and 650 trailers. The thresholds proposed (“over 200 trailers” for BLE adoption and “50–100 trailers” for pilot deployment) are derived from the observed operational and financial outcomes of this

case but should be treated as tentative recommendations rather than universally applicable thresholds. A simple break-even analysis based on the case data suggests that for fleets with more than 200 trailers, the annual savings from error prevention (€19,750 for 650 trailers) exceed the hardware and maintenance costs of BLE beacons, making the investment economically viable. For smaller fleets, the payback period may be longer, and a case-by-case assessment is recommended. A pilot scale of 50–100 trailers is suggested as a practical minimum to observe measurable improvements while minimizing implementation risk. Further research across different fleet sizes and operational contexts is needed to validate and refine these thresholds.

5.6. Link to the Award

The company that was the subject of this case study received the “Transport Company of the Year 2025” award at the Plovdiv Business Awards. According to the official jury statement, the award was given for the “systematic approach to digital transformation, including the deployment of BLE/LTE tracking and TMS–GPS platform integration, which led to measurable improvements in efficiency and operational cost reduction”. This external validation confirms that the innovations not only deliver quantitative ROI but also contribute to the company’s competitive positioning in the industry.

6. Research Contribution

This study makes three main contributions: theoretical, empirical, and practical.

Theoretical contribution: The primary theoretical contribution is the empirical demonstration that error prevention (correct trailer attachment) generates higher returns than asset tracking. While the existing literature [30,31] conceptualizes IoT tracking primarily as a tool for visibility and asset recovery, our results show that for fleets with high trailer swap frequency (≥ 100 per week), the proactive BLE/LTE approach yields significantly higher economic benefits. This finding enriches quality cost theory [14,15] in the logistics context, demonstrating that prevention costs are not only lower than failure costs but also yield measurable and rapid returns.

Empirical contribution: For the first time, quantitative data on the economic impact of BLE/LTE architecture in a fleet of this scale—over 800 tractors and 650 trailers—are published. The presented KPIs (90% excess mileage reduction, 99.9% faster swap registration, 334% ROI over 5 years) provide concrete benchmarks for other companies considering similar investments.

Practical contribution: This study provides clear guidance for fleet managers: (1) BLE beacons are more cost-effective than active GPS trackers for fleets with over 200 trailers; (2) two-way integration between the TMS and GPS platform is critical for achieving maximum effect; (3) a pilot deployment with 50–100 trailers is recommended before scaling.

7. Limitations of This Study

Despite the compelling results, this study has several limitations that should be considered when interpreting the findings.

Lack of specialized temperature-controlled transport: The present case study covers only dry cargo (no refrigerated trailers were used). This limitation does not allow evaluation of BLE sensor effectiveness for real-time temperature monitoring, a functionality that would be critical for the cold chain. Future research could extend the scope by including refrigerated trailers and temperature sensors.

Dependence on LTE/5G coverage: In areas with limited LTE/5G coverage (e.g., remote areas or tunnels), the system may lose connectivity or operate with higher latency. Although the trackers support LTE Cat 20 as fallback, this may affect real-time functionality. It should be noted, however, that this limitation is temporary; with the impending shutdown of 2G networks in the European Union (scheduled for 2026–2028) [32] and the widespread transition to LTE and 5G infrastructure, coverage and network reliability will continue to improve significantly. This will make such real-time applications increasingly stable and accessible even in remote areas and during cross-border operations.

8. Future Research

The results of this study outline several directions for future work that could expand the application of BLE technology in logistics.

Expanding the BLE ecosystem with temperature sensors: Future research could include BLE sensors for measuring ambient temperature in trailers. This would be critical for the cold chain (refrigerated goods, pharmaceuticals, food), where maintaining a constant temperature is mandatory. Some sensors offer battery life of up to 3 years and can operate at extreme temperatures down to $-30\text{ }^{\circ}\text{C}$. This would allow dispatchers to monitor cargo conditions in real time and take action in case of deviations, without requiring expensive equipment on every trailer.

Integration of tire pressure and temperature sensors (TPMSs): Future studies could explore the integration of BLE sensors into tire pressure and temperature monitoring systems. This would have a direct impact on fuel consumption (proper tire pressure can reduce consumption by up to 3–5%) [28], as well as safety: early detection of anomalies could prevent tire blowouts on active traffic lanes, which lead to costly roadside assistance, time loss, and traffic accident risks.

Expanding the sensor ecosystem: BLE technology allows adding sensors for humidity, door opening, shock/vibration and other parameters, enabling comprehensive monitoring of trailer and cargo condition. All these data can be transmitted through the same LTE/5G trackers in tractors, without additional connectivity or SIM cards for each trailer.

Combining BLE with UWB for precise localization: Future research could explore the possibility of combining BLE with UWB (ultra-wideband) technology for precise localization of trailers in yards and terminals. This would enable trailer identification without physical attachment, expanding the system's application beyond its current scope.

Validation in different contexts: Replication of this study in different geographic regions, with different fleet sizes, and in different logistics segments (e.g., refrigerated transport, freight forwarding) is recommended to confirm the external validity of the results.

9. Conclusions

This empirical study demonstrates that integrating BLE sensors, LTE trackers, and a two-way TMS–GPS platform (Kitin) leads to significant operational and financial benefits for a large fleet. The company, which also received the “Transport Company of the Year 2025” award [33], shows how investments in innovation can yield both direct ROI and enhanced reputation and competitiveness.

Key findings:

- 90% reduction in excess mileage (from 10,000 km to 1000 km annually).
- 99.9% faster trailer swap registration (from 5–10 min to under 1 s).
- 90% reduction in wrong attachments.

- Annual savings of approximately €19,750.

The payback period is approximately 14 months (13.8 months) and 334% ROI over 5 years. It is important to note that the recommendations and thresholds presented are derived from a single case study and should be interpreted with caution when applied to fleets with different operational characteristics, fleet sizes, or geographic contexts. Further research is needed to validate the generalizability of these findings.

Scientific contribution: The proposition that error prevention dominates asset tracking is supported by the case study evidence and a comparative cost–benefit analysis presented in Table 10. Even under conservative assumptions, the BLE/LTE system delivers a positive net annual benefit of €18,425, while active GPS tracking generates a net loss when operational costs exceed recovery value. This finding enriches quality cost theory in the logistics context and provides an empirical and economic basis for rethinking IoT strategies in fleets. This study does not claim to have experimentally proven the superiority of prevention over tracking through a controlled comparison but rather demonstrates that prevention yields measurable and sustainable returns that exceed the net value typically associated with asset recovery.

Table 10. Comparative cost–benefit analysis of BLE/LTE vs. active GPS tracking.

Parameter	BLE/LTE System	Active GPS Tracking (Market Estimate)
Annual hardware + maintenance cost	€1325	€78,000–€351,000
Annual value of error prevention	€19,750	Not applicable
Annual value of asset recovery	Not applicable	€32,500–€65,000
Net annual benefit	€18,425	(–€45,500)–(–€286,000)

Recommendations for managers (based on the findings of this case study):

1. For fleets with more than 200 trailers, BLE beacons appear more cost-effective than active GPS trackers, based on the observed savings and break-even analysis from this case. However, this threshold should be validated through a site-specific cost–benefit assessment.
2. Ensure two-way integration between the TMS and GPS platform. A one-way data flow (GPS → TMS) provides visibility but does not prevent routing errors.
3. A pilot deployment with 50–100 trailers is recommended to validate results in the specific operational context of the company before full-scale implementation.
4. Consider expanding the BLE ecosystem with temperature sensors and TPMSs for additional savings and improved safety, subject to a separate cost–benefit analysis for those specific use cases.

Companies that transition from reactive to proactive management can expect not only rapid financial returns but also significant operational KPI improvements and reputational benefits, as demonstrated by the “Transport Company of the Year” award [33].

Author Contributions: R.G.M.: investigation, conceptualization, methodology, software, visualization, and writing—review and editing; S.H.E.: supervision, validation, and writing—review and editing; V.N.M.: data curation and formal analysis. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Kitin.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments: We thank Kitin (truck.kitin.at) for financial support and documentation. We also thank the anonymous Bulgarian logistics company for providing data and for permission to cite the award.

Conflicts of Interest: This research was funded by Kitin. The funding sponsors had no role in the design of this study; in the collection, analyses, or interpretation of data; in the writing of this manuscript; or in the decision to publish the results. The authors declare no conflicts of interest.

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